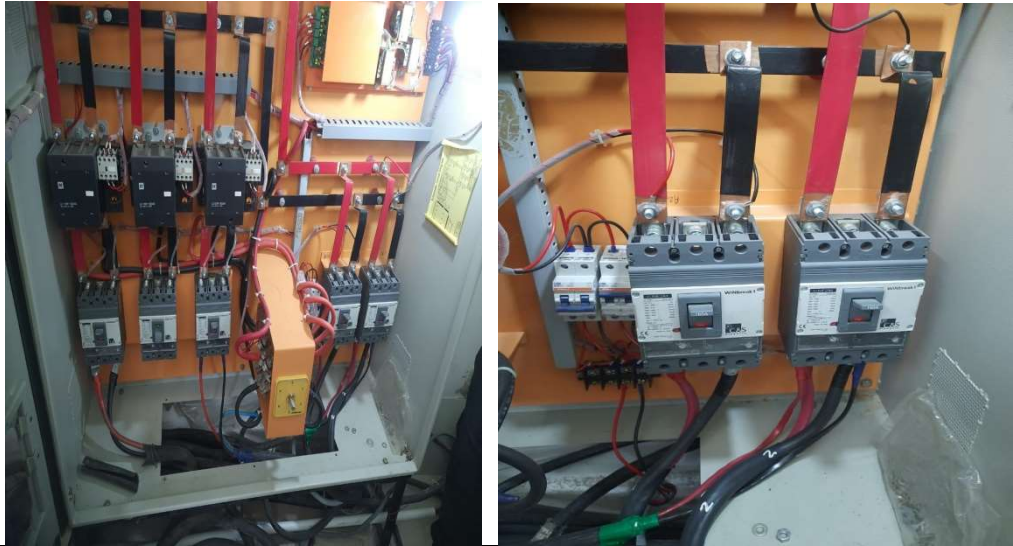


SEC Railway

Inspection of "SAL" station conducted by Pankaj sharma, Sr.DSTE/NGP in connection with 3rd line work on 11/10/2022

Sr.	Observations	Complied by
1	In 3 rd line work, station is to be commissioned with MLK-II Electronic Interlocking by S&T Con/NGP unit. Total nos. of MLK-II Rack – 06 nos. (Center – 02 nos., East - 02 nos., West – 02 nos.)	Dy.CSTE/C/NGP
2	"SAL" station to be commissioned with complete working of 3 rd line signalling gears like Track circuits, Signals, MSDAC etc.	Dy.CSTE/C/NGP
3	CRS sanction is not received for commissioning of "SAL" station in connection with 3 rd line.	Dy.CSTE/C/NGP
4	Approval of "TSAA" yet to be obtained.	Dy.CSTE/C/NGP
5.	Free Home signal to be provided in separate table not on VDU table.	Dy.CSTE/C/NGP
6.	EI History sheet to be prepared including Card status (NEW/Old/Released) and RDSO IC number & date also to be included.	Dy.CSTE/C/NGP
7.	Automatic fire detection & Alarm system is provided. To be incorporated in data-Logger for Remote monitoring.	Dy.CSTE/C/NGP
8.	ELD with Calibration to be provided and to be incorporated in Data Logger.	Dy.CSTE/C/NGP
9.	Tools of IPS, L.C.Gate, Point Machine, MSDAC etc which is supplied along with equipment to be provided to the SSE/Sig/MIB for maintenance of the gear.	Dy.CSTE/C/NGP
10.	Joint testing of Cable meggering is in progress and to be completed before NI.	Dy.CSTE/C/NGP DSTE-I/NGP
11.	Earth value of Location & signals has not checked. Joint testing of Earth Value to be done and completed before NI.	
12.	Approved & updated indoor/outdoor drawings Cable route, Cable Corage, Track Bonding plan etc to provided at station for maintenance after commissioning.	Dy.CSTE/C/NGP
13.	Working of standby MSDAC to be checked jointly with Con unit. Reset Circuit of MSDAC also to be checked. Con unit to be planed for Joint checking of the same.	Dy.CSTE/C/NGP DSTE-I/NGP
14.	Load sharing of DC-DC converter module also to be checked jointly.	
15.	Pre commissioning checklist of MSDAC, IPS, Data Logger, ELD, EI etc to be checked jointly with OEM & open line. Record to be kept at station.	
16.	Data Logger Validation to be done jointly and report to be kept at station.	
17.	Asset Register to be prepared to be kept at the station.	Dy.CSTE/C/NGP
18.	Wiring of potential free contact of Track charger to be done and to be incorporated in Data logger.	Dy.CSTE/C/NGP
19.	Auto changeover CLS Panel having only two output MCB's (160 Amps)). At least 03 no. of MCB to be provided for Centre, East & West.	CEE/Con/NGP
	Image 	
20.	Type of circuits of wires/cables, which are laid through the trough in the Relay room, is to be written over the troughs.	Dy.CSTE/C/NGP

Sr.	Observations	Complied by
21		
22	Old type MSDAC Rack has been used for EC of MSDAC instead of pre-wired Rack. As not to repeat problem faced in KWN, Pre-wired MSDAC with new EC to be provided in all the new installations.	Dy.CSTE/C/NGP
		
23	Compliance of following items to be ensured as per “Guidelines regarding TSAA for Electronic Interlocking system” vide STS/E/TAN/3012, Dt. 28/06/2021.	Dy.CSTE/C/NGP
	Power supply arrangement:	
2 (d)	The location of DC-DC converter shall be made in such a way that the 110V DC power supplies wires do not run and cut across other cables in the Electronic Interlocking room and this cable shall be treated as dirty cable and segregated from other clean wiring.	
	Wiring practices:	
(a).	The input, output, data and power supply cable shall be routed in different cable troughs separated with a gap of minimum 6 to 8 inches.	
(f).	Separation of 110V DC and other external power supply from the internal 12V/24V/50V DC power supply in the relay room by placing them in separate trough/ladder.	
(g).	Voltage monitoring cables to the data logger shall also be placed in a different trough which does not have any internal power supplies.	
24	Cover of Tag blocks which are used for MSDAC, Data logger etc. is to be provided.	Dy.CSTE/C/NGP
25	Inter distance between Relay Racks (trough to ladder of Rack) is found only 80mm. Distance to be increased up to 150 mm for maintenance point of view.	Dy.CSTE/C/NGP
	Image 	

Sr.	Observations	Complied by
26	In front of SM Panel room, Railing not provided at boundary of terrace due to Engg. work. This is very unsafe for S&T & Operating staff working at Central building.	
	image 	
27	Counter of “A” type SPD is provided at front wall of East & west goomty and near stairs at Centre. Same to be provided inside of IPS room just like earlier stations.	
		
28	IPS:	
	Power rack is to be installed near by the FRBC (SMR) to minimise the cable of input supply 230VAC from SPD to IPS.	
	At Centre & East goomty, AMARA RAJA make IPS is provided whereas “HBL” make IPS is provided at West Goomty. There should be only one type of equipment in a station.	
	Spare battery set of IPS to be provided at all locations.	
	At Centre & West, 1 st cycle of charging/discharging of IPS Batteries is completed. At East, discharging of 1 st cycle is in progress.	
	Exhaust Fan is not provided at IPS Room and Battery room of East & west Goomty. Exhaust Fan with wire mesh to provided at Centre, East & west Goomty.	
	Hydrometers stand to be provided at Battery room of Centre, East & west Goomty.	
	IPS ASM monitoring panel of East & west goomty is also to be provided at Centre.	

Sr.	Observations	Complied by
	East & West yard:	
29	Earthing is not connected in any of the Location boxes, signals etc. Earth enclosure is also not provided.	Dy.CSTE/C/NGP
	 	
30	Soldering of Earthing in the location boxes is in progress.	Dy.CSTE/C/NGP
31	Writing work in the Location boxes is in progress.	
32	Sand filling & Plastering at inside the location boxes is in progress.	
33	Distance between Dn. Advance str Signal S-88 and OHE mast is found only 6 meters.	CEE/Con/NGP
		
34	Installation of New Point no. 101A & 153A on ground (3 rd line) is not completed. Also staging of point no. 101B & 153B is not done.	CE/Con/NGP Dy.CSTE/C/NGP
	 	

Sr.	Observations	Complied by
35.	T-arm type insulated SSD to be provided on TWS to avoid failure of Track Circuit due to shorting by SSD.	CE/Con/NGP Dy.CSTE/C/NGP
36	Height of location box is more. Foundation of location box near is completely above the ground level. It may be tilt or fall down due to cutting of soil during rain. Protection of foundation to be provided with concreting around the LB. Steps also to be provided for maintenance.	Dy.CSTE/C/NGP
37	Trench and cables is opened at many places at East & west yard. OFC cable & quad cable is open condition below the East Goomty. Same to be covered by provision of Half round RCC pipe with concreting.  	Dy.CSTE/C/NGP
	 	
38	3rd line formation at "SAL" west yard is in progress.  	CE/Con/NGP

Sr.	Observations	Complied by
39	GI Pipes is provided at Centre, East & West goomty for cable entries in the relay room. Same to be protected by brick wall around the pipe. This will also protect cables from the miscreant's activities. 	Dy.CSTE/C/NGP
41	Pt. No. 41B: checked & Tested Point with 5 mm & 1.6 mm obstruction test piece and found respond correctly. LHTR Chipped off, Out of square about 20mm. Last maintenance of point was done on 24/09/2022 by SSE/Sig/KP. 	Sr.DEN/C/NGP DSTE-I/NGP

Sr.	Observations	Complied by																														
42	Pt. No. 33B: checked & Tested Point with 5 mm & 1.6 mm obstruction test piece and found respond correctly. <table><tr><th colspan="2">Opening (mm)</th><th colspan="2">Housing (Sleeper)</th><th colspan="2">Operating voltage</th><th colspan="2">Operating current</th><th colspan="2">Obstruction vol. &current</th></tr><tr><th>LH</th><th>RH</th><th>LH</th><th>RH</th><th>N – R</th><th>R – N</th><th>N – R</th><th>R – N</th><th>N – R</th><th>R – N</th></tr><tr><td>115</td><td>110</td><td>5 SL</td><td>5 SL</td><td>105 V</td><td>105 V</td><td>1.8 A</td><td>1.6 A</td><td>84 V 3.3A</td><td>81 V 3.1 A</td></tr></table> Last maintenance of point was done on 24/09/2022 by SSE/Sig/KP. Point machine is to be replaced due to manufacturing year is 2003 and codal life exhausted.	Opening (mm)		Housing (Sleeper)		Operating voltage		Operating current		Obstruction vol. ¤t		LH	RH	LH	RH	N – R	R – N	N – R	R – N	N – R	R – N	115	110	5 SL	5 SL	105 V	105 V	1.8 A	1.6 A	84 V 3.3A	81 V 3.1 A	Dy.CSTE/C/NGP DSTE-I/NGP
Opening (mm)		Housing (Sleeper)		Operating voltage		Operating current		Obstruction vol. ¤t																								
LH	RH	LH	RH	N – R	R – N	N – R	R – N	N – R	R – N																							
115	110	5 SL	5 SL	105 V	105 V	1.8 A	1.6 A	84 V 3.3A	81 V 3.1 A																							
43	230V/110V/1KVA transformer to be provided at East & West L. C. Gate Goomty for operation of L.C.Gate.	Dy.CSTE/C/NGP																														
44	At Gate goomty cum auto Goomty at LC 558 between KM 1106/26-28 (SAL-KNHN), Auto changeover CLS with UP & DN AT is available but power supply for Gate and Auto circuits is extending from West goomty. Auto changeover CLS panel is to provide at new gate Lodge and existing UP & DN AT to be utilised along with provision of Mini IPS for power supply at Gate and Auto circuits.	CEE/Con/NGP																														
45	Input supply of ELD to be provided from spare signal transformer which is fed through Inverter 230/110VAC.	Dy.CSTE/C/NGP																														
46	MSDAC DP DA11/DB11 of Dn advance is to be shifted in connection with 3rd line work. Location wiring is not done and drilling of hole in rail also not done for provision of DP. 	Dy.CSTE/C/NGP																														

Pankaj Sharma

Pankaj Sharma
Sr.DSTE/NGP/SECR

1. PS to DRM/NGP for kind information of DRM please.
2. CSE/SECR/BSP for kind information please.
3. CSTE/Con/SECR/BSP for kind information please.
4. CE/Con/NGP for kind information please.
5. CEE/Con/NGP for kind information please.
6. Dy.CSTE/Con/NGP for kind information please